CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

VIEIRA LIVESTOCK

3800 S CENTRAL AVE

TURLOCK CA 95380

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vieira Livestock

Physical address of dairy:

3800 S Central AVE	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 08/15/1968

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X028-X017-XXXX

B. OPERATORS

Vieira, Joe & Christine

Operator name: Vieira, Joe & Christine

Telephone no.: (209) 531-4156 (Landline) (Cellular)

3800 Central AVE	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Vieira, Joe & Christine

Legal owner name: Vieira, Joe & Christine

Telephone no.: (209) 531-4156 (Landline) (Cellular)

3800 Central AVE	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	20	45	39	46
Number under roof	0	20	0	0	0	0
Maximum number	0	20	20	45	39	46
Average number	0	20	20	45	39	46
Avg live weight (lbs)	0	1,250	875	550		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,175.64 *tons per reporting period*

Total nitrogen from manure: 14,162.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 9,913.40 *lbs per reporting period*

Total phosphorus from manure: 1,832.85 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 4,599.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 200,000 *gallons*

Total nitrogen generated: 310.77 *lbs*

Total phosphorus generated: 100.31 *lbs*

Total potassium generated: 5,190.59 *lbs*

Total salt generated: 28,706.80 *lbs*

	200,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	200,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	10	10	2	process wastewater	X058-X028-X017-XXXX
Field 2 (Mitchell)	15	15	1	none	X058-X029-X017-XXXX
Totals for areas that were used for application	10	10	2		
Totals for areas that were not used for application	15	15	1		
Land application area totals	25	25	3		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/28/2022: Oats, hay

Crop: Oats, hay Acres planted: 10 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	34.00 *ton*	Dry-weight		10.7	24,700.00	3,400.00	35,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	3.40	149.99	20.65	216.18	619.38

06/19/2023: Sudangrass, hay

Crop: Sudangrass, hay Acres planted: 10 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	38.00 *ton*	Dry-weight		5.8	10,900.00	3,200.00	32,000.00		12.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	128.00	17.60	132.00	0.00
Total actual harvest content	3.80	78.04	22.91	229.09	928.55

Field 2 (Mitchell)

Field 2 (Mitchell)

Field name: Field 2 (Mitchell)

11/26/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 15 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/20/2023	107.00 *ton*	Dry-weight		6.3	21,100.00	2,000.00	19,600.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.13	282.06	26.74	262.01	1,149.64

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/28/2022: Oats, hay

Field name: Field 1

Crop: Oats, hay — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.08	0.00	0.00	0.00	1,699,243.00 *gal*
Application event totals		9.08	0.00	0.00	0.00	

Field 1 - 06/19/2023: Sudangrass, hay

Field name: Field 1

Crop: Sudangrass, hay — Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.73	0.00	0.00	0.00	2,946,136.00 *gal*
Application event totals		15.73	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.75	0.00	0.00	0.00	2,575,262.00 *gal*
Application event totals		13.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.08	10.03	519.06	2,870.68	200,000.00 *gal*
Canal	Surface water	12.49	0.00	0.00	0.00	2,337,876.00 *gal*
Application event totals		43.56	10.03	519.06	2,870.68	

Field 1 - 06/19/2023: Sudangrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	12.88	0.00	0.00	0.00	2,411,334.00 *gal*
Application event totals		12.88	0.00	0.00	0.00	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.31	0.00	0.00	0.00	2,678,507.00 *gal*
Application event totals		14.31	0.00	0.00	0.00	

Field 2 (Mitchell) - 11/26/2022: Alfalfa, hay

Field name: Field 2 (Mitchell)

Crop: Alfalfa, hay

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.67	0.00	0.00	0.00	4,120,875.00 *gal*
Application event totals		14.67	0.00	0.00	0.00	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	16.13	0.00	0.00	0.00	4,530,906.00 *gal*
Application event totals		16.13	0.00	0.00	0.00	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.21	0.00	0.00	0.00	3,708,807.00 *gal*
Application event totals		13.21	0.00	0.00	0.00	

Field 2 (Mitchell) - 11/26/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.14	0.00	0.00	0.00	4,252,751.00 *gal*
Application event totals		15.14	0.00	0.00	0.00	

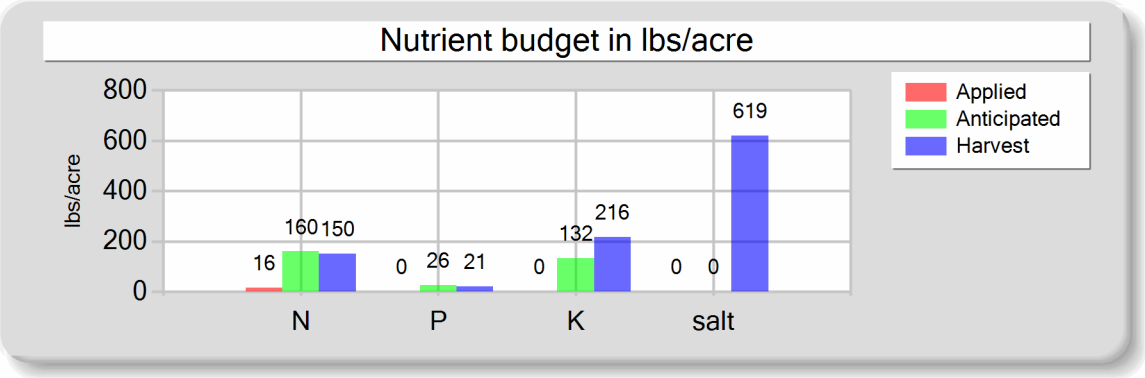
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.73	0.00	0.00	0.00	3,294,653.00 *gal*
Application event totals		11.73	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field 1 - 10/28/2022: Oats, hay

Field name: Field 1 Crop: Oats, hay Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.08	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	16.08	0.00	0.00	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	149.99	20.65	216.18	619.38
Nutrient balance	-133.91	-20.65	-216.18	-619.38
Applied to removed ratio	0.11	0.00	0.00	0.00

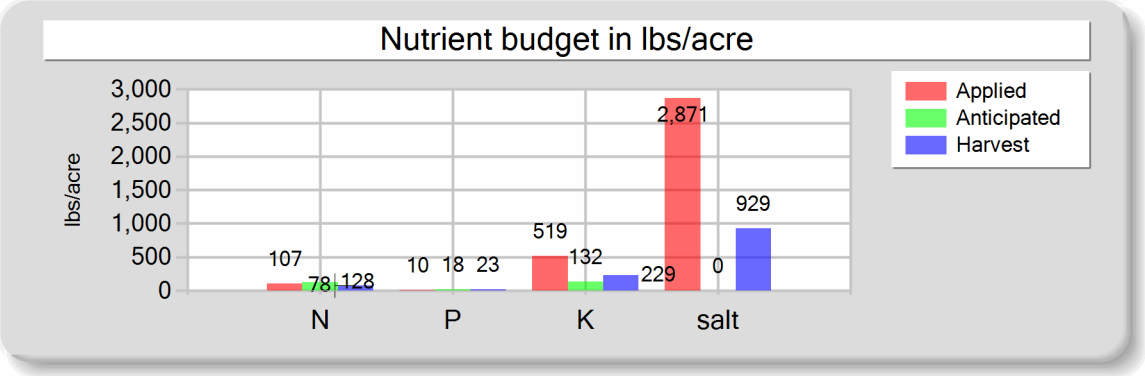
Fresh water applied	
1,699,243.00	*gallons*
62.58	*acre-inches*
6.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/19/2023: Sudangrass, hay

Field name: Field 1 Crop: Sudangrass, hay Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	31.08	10.03	519.06	2,870.68
Fresh water	69.16	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.24	10.03	519.06	2,870.68
Anticipated crop nutrient removal	128.00	17.60	132.00	0.00
Actual crop nutrient removal	78.04	22.91	229.09	928.55
Nutrient balance	29.20	-12.88	289.96	1,942.13
Applied to removed ratio	1.37	0.44	2.27	3.09

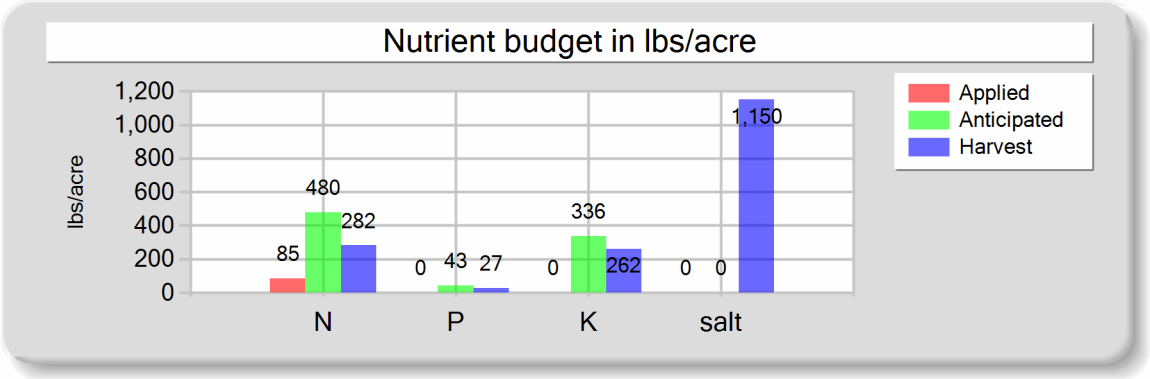
Fresh water applied	
12,949,115.00	*gallons*
476.87	*acre-inches*
47.69	*inches/acre*

Process wastewater applied	
200,000.00	*gallons*
7.37	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 (Mitchell) - 11/26/2022: Alfalfa, hay

Field name: Field 2 (Mitchell) Crop: Alfalfa, hay Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	70.88	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	84.88	0.00	0.00	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	282.06	26.74	262.01	1,149.64
Nutrient balance	-197.18	-26.74	-262.01	-1,149.64
Applied to removed ratio	0.30	0.00	0.00	0.00

Fresh water applied	
19,907,992.00	*gallons*
733.14	*acre-inches*
48.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 9.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	185.00	109.00	0.00	1.20	60.10	3,110.00	50.50	114.00	2,100.00	3,350.00	726.00	1,690.00	2,140.00	19,100.00	17,200
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	6.40								284.00	0
DL		0.01	0.01								0.01	1

Domestic Well

Domestic Well

DW House

Sample description: DW House

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.40	0.00	14.40								1,910.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/28/2022: Oats, hay

Field 1 Oat Hay

Sample and source description: Field 1 Oat Hay

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	3,400.00	35,600.00		10.20
DL	0.01	0.01	0.01		1.00

Field 1 - 06/19/2023: Sudangrass, hay

Field 1 - 06/19/2023: Sudangrass, hay

Field 1 Sudan Hay

Sample and source description: Field 1 Sudan Hay

Sample date: 09/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	3,200.00	32,000.00		12.97
DL	0.01	0.01	0.01		1.00

Field 2 (Mitchell) - 11/26/2022: Alfalfa, hay

Field Mitchell Alfalfa

Sample and source description: Field Mitchell Alfalfa

Sample date: 08/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	2,000.00	19,600.00		8.60
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

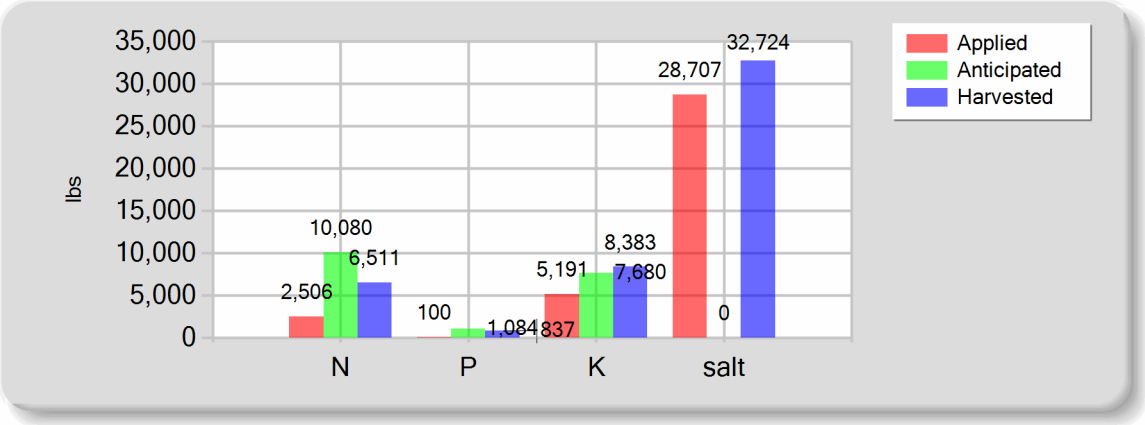
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

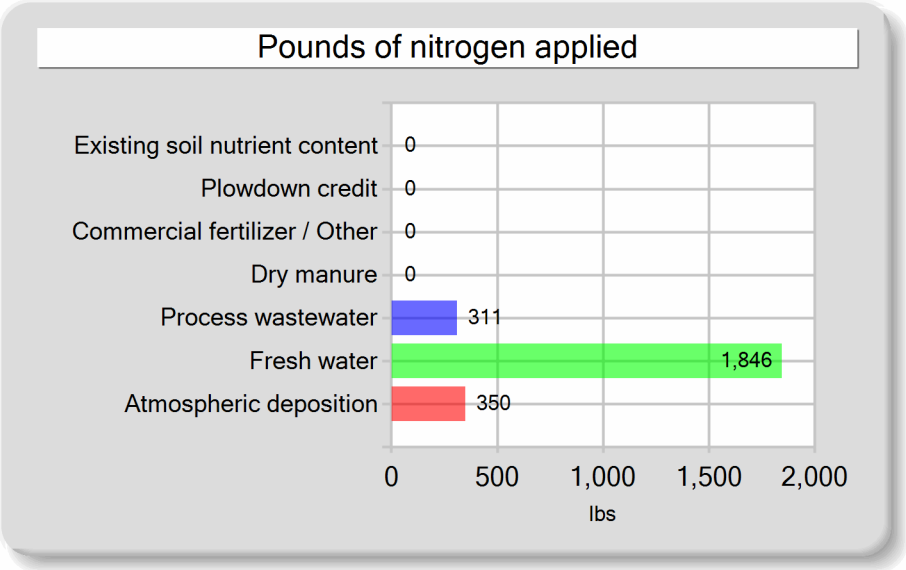
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

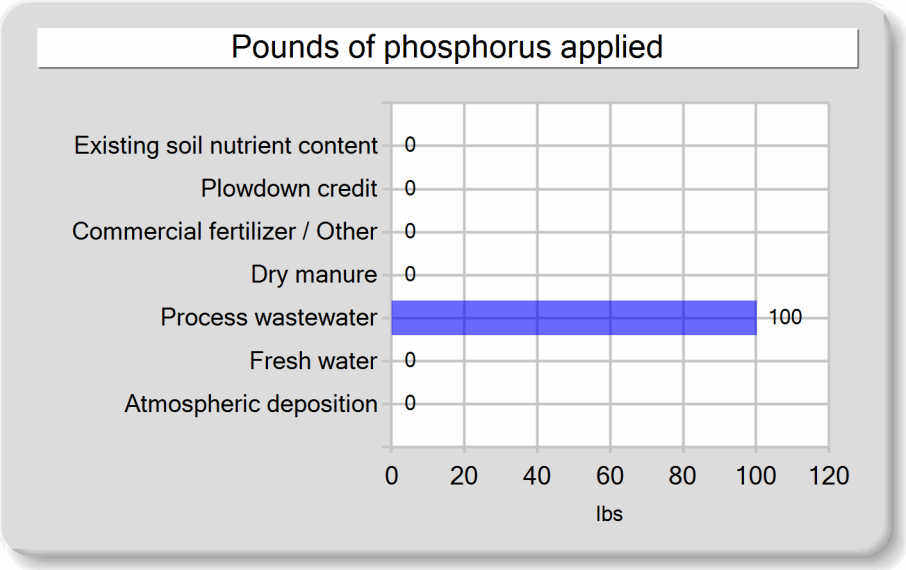
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	310.77	100.31	5,190.59	28,706.80
Fresh water	1,845.59	0.00	0.00	0.00
Atmospheric deposition	350.00	0.00	0.00	0.00
Total nutrients applied	2,506.35	100.31	5,190.59	28,706.80
Anticipated crop nutrient removal	10,080.00	1,084.00	7,680.00	0.00
Actual crop nutrient removal	6,511.17	836.59	8,382.87	32,723.88
Nutrient balance	-4,004.81	-736.29	-3,192.28	-4,017.08
Applied to removed ratio	0.38	0.12	0.62	0.88

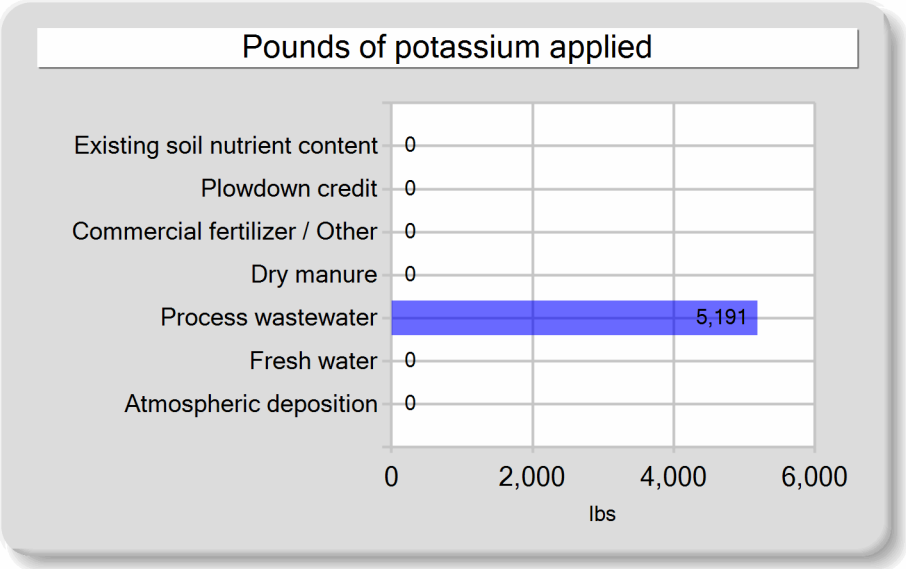
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

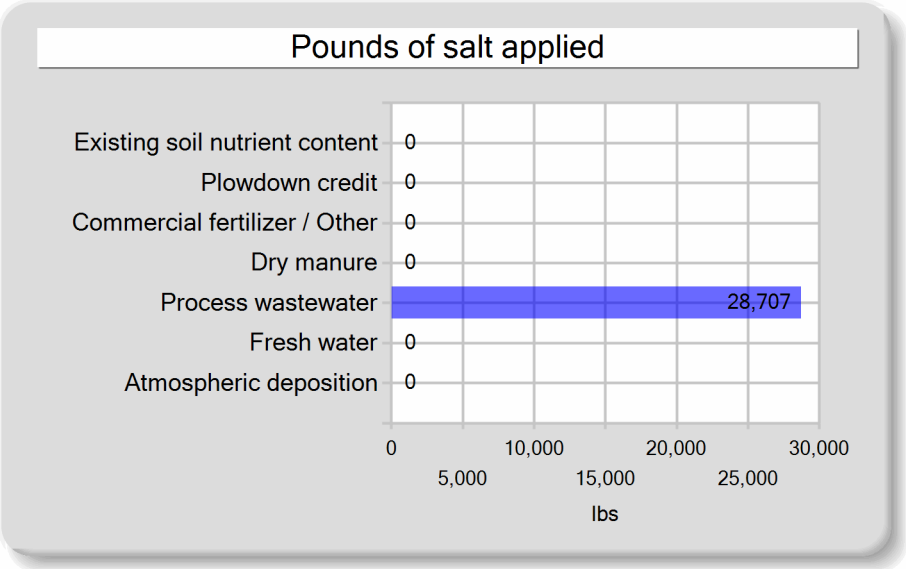


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No wastewater generated as no milking activities are occurring onsite, therefore wastewater is minimal and minimal has been applied.
Also, minimal manure is produced due to the low number of animals housed onsite, therefore no manure activity occurred nor any samples taken

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe & Christine Vieira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0704
Sampled By: M.Dekeyrel
Requested By: Joe & Christine Vieira
Submitted Date: 12/12/23
Reported Date: 12/18/23
Project:
Crop ID:

Vieira Livestock
3800 S Central Ave
Turlock CA 95380
00-0021380 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - House	12/11/23	16:20	1910	1.91	14.4	ND		7.4

***** = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland

Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0704
Sampled By: M.Dekeyrel
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Project:
Crop ID:

Vieira Livestock
3800 S Central Ave
Turlock CA 95380
00-0021380 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	DW - House	12/11/23	16:20	**1910**	**1.91**	**14.4**	39.31	3.28

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Vieira Livestock
3800 S Central Ave
Turlock, CA 95380

Account# 00-0021380
Account Manager: Ben Nydam
Submitted By: Joe & Christine Vieira

Received: 12/12/2023 15:55
Reported: 12/18/2023 09:18

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0704-01	DW - House	Well Water	M.Dekeyrel		12/11/2023 16:20

Default Cooler Temperature on Receipt °C: -1.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Vieira Livestock
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Turlock, CA 95380

Account# 00-0021380
Account Manager: Ben Nydam
Submitted By: Joe & Christine Vieira

Received: 12/12/2023 15:55
Reported: 12/18/2023 09:18

Sample Results

Sample: DW - House
23L0704-01 (Water)

Sampled: 12/11/2023 16:20
Sampled By: M.Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.91**	mmhos/cm	0.01	1		12/13/23 17:28	SM 2510 B		BEL0497
Electrical Conductivity umhos	**1910**	umhos/cm	10.0	1		12/13/23 17:28	SM 2510 B		BEL0497
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 14:20	Field		BEL0390
Nitrate Nitrogen as NO3N	**14.4**	mg/L	0.1	1	10	12/13/23 13:03	EPA 300.0		BEL0351
Temperature	**25.0**	units	0.0	1		12/13/23 17:28	SM 4500-H+	H	BEL0497
pH	**7.4**	units	1.0	1		12/13/23 17:28	SM 4500-H+	H	BEL0497

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DELLAVALLE™ LABORATORY INC

Vieira Livestock
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Turlock, CA 95380

Account# 00-0021380
Account Manager: Ben Nydam
Submitted By: Joe & Christine Vieira

Received: 12/12/2023 15:55
Reported: 12/18/2023 09:18

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0351										
Blank (BEL0351-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0351-BLK5)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0351-BS1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
LCS (BEL0351-BS2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0351-BS3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEL0351-BS4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
Duplicate (BEL0351-DUP1)		**Source: 23L0613-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	1.8		0.1	mg/L		1.8			0.900	10
Duplicate (BEL0351-DUP2)		**Source: 23L0664-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.7		0.1	mg/L		6.7			0.0448	10
Duplicate (BEL0351-DUP3)		**Source: 23L0695-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.1		0.1	mg/L		4.1			0.441	10
Duplicate (BEL0351-DUP4)		**Source: 23L0718-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			1.70	10
Matrix Spike (BEL0351-MS1)		**Source: 23L0613-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	7.1		0.1	mg/L	5.000	1.8	106	90-110		
Matrix Spike (BEL0351-MS2)		**Source: 23L0664-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	11.7		0.1	mg/L	5.000	6.7	101	90-110		

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3800 S Central Ave
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Account Manager: Ben Nydam
Submitted By: Joe & Christine Vieira

Received: 12/12/2023 15:55
Reported: 12/18/2023 09:18

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0351 (Continued)										
Matrix Spike (BEL0351-MS3)			**Source: 23L0695-01**		Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3		0.1	mg/L	5.000	4.1	105	90-110		
Matrix Spike (BEL0351-MS4)			**Source: 23L0718-02**		Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.6		0.1	mg/L	5.000	1.3	105	90-110		
Reference (BEL0351-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0351-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0351-SRM3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0351-SRM4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0351-SRM5)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0497										
Blank (BEL0497-BLK1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0497-BLK2)					Prepared & Analyzed: 12/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0497-BLK3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.2		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0497-DUP1)		**Source: 23L0694-04**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1.14		0.01	mmhos/cm		1.13			0.986	10
pH	7.8		1.0	units		7.8			0.129	10
Electrical Conductivity umhos	1140		10.0	umhos/cm		1130			0.986	10
Duplicate (BEL0497-DUP2)		**Source: 23L0704-01**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1.94		0.01	mmhos/cm		1.91			1.40	10
pH	7.3		1.0	units		7.4			0.815	10
Electrical Conductivity umhos	1940		10.0	umhos/cm		1910			1.40	10
Reference (BEL0497-SRM1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	456			umhos/cm	426.0		107	90-110		
Reference (BEL0497-SRM2)					Prepared & Analyzed: 12/13/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0497-SRM3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0497-SRM4)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0497-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		

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LABORATORY INC

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3800 S Central Ave
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Account# 00-0021380
Account Manager: Ben Nydam
Submitted By: Joe & Christine Vieira

Received: 12/12/2023 15:55
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0497 (Continued)										
Reference (BEL0497-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0497-SRM6)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0497-SRM7)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0497-SRM8)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/12/23 15:55 23L0704

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DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21380 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: **Vieira Livestock**
Address: 3800 S Central Ave
City, State, Zip: Turlock, CA 95380
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Joe & Christine Vieira - (209) 531-4156

Ranch _____

Date sampled: 12/11/23

Sampled by: MDelgado

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. Dairy House	Sampled From:	12/11/23	420p	Ø	-1.2
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDelgado	CASInc	12/11/23 420p	12/12/23 @9:30am
Second	JLF	DLV	12-12-23 9:30	12-12-23 2:01
Third	MM	DCI	12/12 15:55	
Fourth				

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/12/23 15:55 23L0704

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1									
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1									
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT